डॉ.श्यामाप्रसाद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान
DR.SHYAMA PRASAD MUKHERJEE INTERNATIONAL
INSTITUTE OF INFORMATION TECHNOLOGY, NAYA RAIPUR
(A Joint Initiative of NTPC & Govt. of Chhattisgarh)

Email: iiitnr@iiitnr.ac.in
Tel: +91-771- 2474040
Web: www.iiitnr.ac.in

M.Tech - ECE- PG curriculum

(With effect from July 2025)

Total Credit : 80

Lecture-L: 1 Credit = 1 hr/ week

Tutorial-T: 1 Credit = 1hr/week

Practical-P: 1 Credit = 2 hr/week

Scholarly Engagement Hours-S: 0 Credit= 2 hr/week

Curriculum

Semester 1 Total Credit : 22					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Optimization Algorithms	EC-L-*xx*-C	Core	3	2–1–0-2
2	VLSI Design	EC-L-*xx*-C	Core	3	2–0–2-2
3	Intelligent Systems	EC-L-*xx*-C	Core	3	2–0–2-2
4	Programming Lab – 1 (FPGA / Python)	EC-L-*xx*-C	Skill	2	1–0–2-2
5	Program Elective I	EC-L-*xx*-E	PEC	3	3-0-0-0
6	Program Elective II	EC-L-*xx*-E	PEC	3	3-0-0-0
7	Program Elective III	EC-L-*xx*-E	PEC	3	3-0-0-0
8	Personality Development and Life Skills	HS-L-*xx*-C	VSEC	2	1-0-2-2
			Total	**22**	**17–1–06** / **24**

Semester 2 Total Credit : 22					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	IOT Networks	EC-L-*xx*-C	Core	3	2–1–0-2
2	Practical Component / Hardware Competitions	EC-L-*xx*-C	Skill	2	1–0–2-2
3	Program Elective IV	EC-L-*xx*-C	PEC	3	3-0-0-0
4	Program Elective V	EC-L-*xx*-C	PEC	3	3-0-0-0
5	Program Elective VI	EC-L-*xx*-E	PEC	3	3-0-0-0
6	Program Elective VII	EC-L-*xx*-E	PEC	3	3-0-0-0
7	Open Elective I	EC-L-*xx*-E	OE	3	3-0-0-0
8	Research Methodology	HS-L-*xx*-C	RM	2	1-0-2-2
			Total	**22**	**19–1–04** / **24**

Summer Semester					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Research Seminar	EC-L-*xx*-C	ELC	2	-
2	Internship / Thesis / Project – Phase I	EC-L-*xx*-C	ELC	2	-
			Total	**04**	**--**

Email: iiitnr@iiitnr.ac.in
Tel: +91-771- 2474040
Web: www.iiitnr.ac.in

Exit: After successful completion of one year (first two semesters) along with Summer Semester, a student may get an exit option as per ordinance and will be eligible to a PGDM Certificate. They need to do submit a Research Paper OR need to carryout an internship in a company as a part of Summer Semester for the award of PGDM certificate.

Semester 3 Total Credit : 16					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Thesis / Project / Internship / Entrepreneurship – Phase II	EC-L-*xx*-C	EL	16	-
			Total	**16**	--

Semester 4 Total Credit : 16					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Thesis / Project / Internship / Entrepreneurship – Phase III	EC-L-*xx*-C	EL	16	-
			Total	**16**	--

Important Instructions

M.Tech (ECE) – Thesis / Capstone Project Policy (As per NEP 2020 Guidelines and Outcome-Based Framework)
Applies to: Semester 3 and Semester 4. Total Credits: Up to 32 credits (16 per semester)

Overview

All students must pursue a Thesis / Capstone Project / Industry Internship spanning Semesters 3 and 4.
Students must choose one of three tracks at the start of Semester 3:

- Research-Oriented Track
- Entrepreneurship-Oriented Track
- Industry Internship Track

Flexibility to Switch Track

Students are permitted to switch from one track to another after completion of Semester 3, subject to:

- Approval of the Academic Advisor / Thesis Committee
- Submission of a revised work plan
- Alignment with credit and evaluation expectations of the new track
- Upon switching, the evaluation in Semester 3 will be based on the original track, while Semester 4 will be evaluated as per the new choice.

Track 1: Research-Oriented

Objective: Advance scientific and applied research.
Semester 3 Deliverables:

- Research proposal, problem formulation
- Initial literature survey and methodology

Semester 4 Deliverables:

- At least one research paper (Tier I/II conference or SCI journal) submitted/communicated

OR

Track 2: Entrepreneurship-Oriented

Objective: Promote innovation and technology-driven entrepreneurship.
Semester 3 Deliverables:

- Problem-solution definition
- Market need validation
- Initial prototype or MVP design

Semester 4 Deliverables:

- Startup prototype developed up to TRL 4–6
- Incubation or pre-incubation involvement

Track 3: Industry Internship

Objective: Develop industry-aligned skills and solve real-world problems.
Semester 3 Deliverables:

- Internship plan and problem statement
- Weekly progress reports / mid-term review

Semester 4 Deliverables:

- Full-time internship (minimum 16 weeks)
- Internship completion certificate

Email: iiitnr@iiitnr.ac.in
Tel: +91-771- 2474040
Web: www.iiitnr.ac.in

• One patent filed Complete thesis and viva	Final report, product demo, and business model	Project report, industry mentor evaluation, and presentation

General Guidelines

- Students will be assigned an academic supervisor and, where applicable, an industry/startup mentor.
- Periodic reviews will be conducted during both semesters to ensure progress and accountability.
- Evaluation will follow track-specific rubrics, based on learning outcomes and final deliverables.
- Interdisciplinary problems, multi-student teams (for startups), and industry-collaboration are encouraged.

Elective Streams (Tentative)

VLSI Design	Intelligent Systems	Advanced Communications
Digital System Design	Signals and Systems for Intelligent Devices	Principles of Digital Communication
CMOS VLSI Design	Digital Signal Processing (DSP)	Wireless Communication Systems
Analog Integrated Circuit Design	Sensors and Instrumentation for Smart Systems	Advanced Signal Processing for Communications
VLSI Testing and Testability	Microcontrollers and Embedded Systems	MIMO and Massive MIMO Systems
VLSI Physical Design Automation	Machine Learning for Embedded Applications	5G Communication Systems
Low Power VLSI Design	Intelligent Control Systems	Millimeter Wave and Terahertz Communications
Design of High-Speed Digital Systems	Hardware-Software Co-design for Intelligent Systems	Cognitive Radio and Dynamic Spectrum Access
Advanced Topics in VLSI (e.g., SoC, NoC, RF IC Design)	Advanced Topics in Intelligent Electronics (e.g., Neuromorphic Computing, Edge AI, Brain–Computer Interface)	6G and Future Wireless Technologies

Electives are subject to change depending upon the availability of expertise, industry collaboration, and academic planning.

The institute retains the flexibility to:

- **Revise or replace elective courses** based on emerging technologies, evolving industry needs, and faculty expertise.
- Offer electives in **collaboration with industry partners**, research labs, or through **online platforms** (e.g., NPTEL, SWAYAM, Coursera) with appropriate academic integration.
- Allow students to **take interdisciplinary electives** from other departments or specializations, with prior approval and after meeting prerequisites.
- Provide opportunities for **guest lectures, micro-courses, workshops**, and **credit-bearing industry projects** in lieu of or alongside elective courses.

All changes will be communicated at the beginning of each semester through the academic office and course planning committee.